

Lecture 3: Wealth and Capital Taxation Revisited: Straub-Werning

Abdoulaye Ndiaye

NYU

December 24, 2025

Straub-Werning (2018)

- ▶ Something is very fishy here
- ▶ Start with Judd V.1, assume CES preferences and $\gamma = 0$
- ▶ Immediately

$$\max_{c, C, k} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + C_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$$

$$\underbrace{(1 - \sigma) \sum_{t=0}^{\infty} \beta^t U(C_t)}_{\downarrow \text{ in } C \text{ if } \sigma > 1} = \underbrace{U'(C_0)}_{\downarrow \text{ in } C} k_0$$

- ▶ There exists sequence $\{C_t\}$ with $C_t \rightarrow 0$ that satisfy the last constraint

Judd Version 1 Revisited

- ▶ In the limit $\{C_t\}_t \rightarrow 0$ we are solving the first best

$$\max_{c, C, k} \sum_{t=0}^{\infty} \beta^t u(c_t)$$

subject to

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t$$

- ▶ Hence $\{C_t\}_t \rightarrow 0$ is optimal, the only feasible way is to set tax $\tau_{kt} = 1$ in some t
- ▶ τ_{kt} is a tax on wealth: $\tau_{kt} = 1 \iff$ full expropriation
- ▶ it is equivalent to an infinite tax on interest income.
- ▶ How about Judd v2 and Chamley (application of Diamond-Mirrlees)?

Judd Version 2

- ▶ Assume $U(C) = C^{1-\sigma}/(1-\sigma)$
- ▶ μ_t = multiplier on IC constraint, $\kappa_t = k_t/C_{t-1}$, $v_t = U'(C_t)/u'(c_t)$

$$\mu_0 = 0$$

$$\mu_{t+1} = \mu_t \left(\frac{\sigma - 1}{\sigma \kappa_{t+1}} + 1 \right) + \frac{1}{\beta \sigma \kappa_{t+1} v_t} (1 - \gamma v_t)$$

$$\frac{u'(c_{t+1})}{u'(c_t)} (f'(k_{t+1}) + 1 - \delta) = \frac{1}{\beta} + v_t (\mu_{t+1} - \mu_t)$$

- ▶ Judd (1985) studies interior steady state
 - ▶ for allocation + multipliers
 - ▶ $c_t = c > 0$, $C_t = C > 0$, $k_t = k > 0$, $\mu_t = \mu$
 - ▶ Last FOC $\Rightarrow R^* = 1/\beta$
 - ▶ Capitalists' Euler $\Rightarrow R = 1/\beta$
 - ▶ Hence: **Zero capital tax!**

Judd Version 2 Revisited

► ... or not ???

- non convergence of allocation (cycles) ?
- convergence to non-interior steady state?
- non convergence of multipliers?

Judd Version 2 Revisited

- ▶ ... or not ???
- ▶ non convergence of allocation (cycles) ?
- ▶ convergence to non-interior steady state?
- ▶ non convergence of multipliers?

Log case

- ▶ Simple special case: $\sigma = 1$, $U(C) = \log C$
 $\Rightarrow$ constant savings rate β ,

$$C_t = (1 - \beta)R_t k_t$$
$$k_{t+1} = \beta R_t k_t = \frac{\beta}{1 - \beta} C_t$$

- ▶ Substitute out C_t in planning problem (with $\gamma = 0$)

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t)$$

$$c_t + \frac{1}{\beta} k_{t+1} + g \leq f(k_t) + (1 - \delta)k_t$$

- ▶ Like a neoclassical growth model, with higher cost of capital!

Log case

- ▶ Converges to unique interior steady state

- ▶ Planner's Euler: $R^* = 1/\beta^2$

- ▶ Capitalists: $R = 1/\beta$

$$\text{tax} = 1 - \frac{R}{R^*} = 1 - \beta$$

- ▶ Why positive tax?

- ▶ multipliers do not converge (Reinhorn 2002)

- ▶ Is this specific to log preferences?

- ▶ Lansing (1999): Yes, "knife-edged"

- ▶ Werning-Straub: **No!** Positive capital taxation for all $\sigma \geq 1$!

Log case

- ▶ Converges to unique interior steady state

- ▶ Planner's Euler: $R^* = 1/\beta^2$

- ▶ Capitalists: $R = 1/\beta$

$$\text{tax} = 1 - \frac{R}{R^*} = 1 - \beta$$

- ▶ Why positive tax?

- ▶ multipliers do not converge (Reinhorn 2002)

- ▶ Is this specific to log preferences?

- ▶ Lansing (1999): Yes, "knife-edged"

- ▶ Werning-Straub: **No!** Positive capital taxation for all $\sigma \geq 1$!

First order conditions

- ▶ Take planning problem FOCs with $\gamma = 0$
- ▶ Suppose allocation did converge to interior steady state (c, C, k)
- ▶ Law of motion for multiplier μ_t of capitalists' IC

$$\mu_0 = 0$$

$$\mu_{t+1} = \underbrace{\mu_t \left(\frac{\sigma - 1}{\sigma \kappa_{t+1}} + 1 \right)}_{\rightarrow \text{const} > 1} + \underbrace{\frac{1}{\beta \sigma \kappa_{t+1} v_t}}_{\rightarrow \text{const} > 0}$$

Hence μ_t diverges to $+\infty$, and so does $\mu_{t+1} - \mu_t$

- ▶ FOC for capital gives a **contradiction**:

$$\underbrace{\frac{u'(c_{t+1})}{u'(c_t)}}_{\rightarrow 1} \underbrace{(f'(k_{t+1}) + 1 - \delta)}_{\rightarrow \text{const}} = \frac{1}{\beta} + \underbrace{v_t(\mu_{t+1} - \mu_t)}_{\rightarrow +\infty}$$

Positive long run capital taxation

- ▶ This proves:
- ▶ **Proposition 2: S-W** If $\sigma > 1$, the optimal allocation **cannot** be converging to the zero capital tax steady state
 - ▶ ... or in fact, *any* other interior steady state
- ▶ Next result shows the system converges to a *non*-interior steady state!
- ▶ **Proposition 3: S-W** If $\sigma > 1$, the optimal allocation satisfies

$$c_t \rightarrow 0 \quad k_t \rightarrow k_g \quad C_t \rightarrow \frac{1-\beta}{\beta} k_g$$

$$\text{tax} = 1 - \frac{R_t}{R_t^*} \rightarrow \mathcal{T}_g > 0$$

where $\mathcal{T}_g \rightarrow 100\%$ as $g \rightarrow 0$.

- ▶ Here k_g is the lowest feasible steady state capital stock, $\frac{1}{\beta} k_g + g = f(k_g) + (1 - \delta)k_g$

Positive long run capital taxation

- ▶ This proves:
- ▶ **Proposition 2: S-W** If $\sigma > 1$, the optimal allocation **cannot** be converging to the zero capital tax steady state
 - ▶ ... or in fact, *any* other interior steady state
- ▶ Next result shows the system converges to a *non*-interior steady state!
- ▶ **Proposition 3: S-W** If $\sigma > 1$, the optimal allocation satisfies

$$c_t \rightarrow 0 \quad k_t \rightarrow k_g \quad C_t \rightarrow \frac{1-\beta}{\beta} k_g$$

$$\text{tax} = 1 - \frac{R_t}{R_t^*} \rightarrow \mathcal{T}_g > 0$$

where $\mathcal{T}_g \rightarrow 100\%$ as $g \rightarrow 0$.

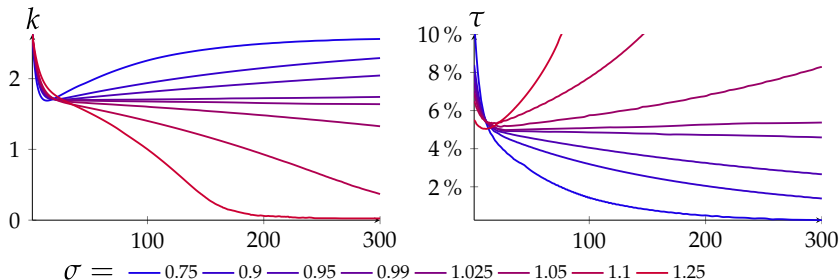
- ▶ Here k_g is the lowest feasible steady state capital stock, $\frac{1}{\beta} k_g + g = f(k_g) + (1 - \delta)k_g$

Intuition

- ▶ Intuition: Incentivizing capitalists' savings behavior through anticipatory effects
 - ▶ Start with a constant tax
 - ▶ Announce a tax increase in the far future
 - ▶ $IES < 1 \Rightarrow$ capitalists increase savings today
 - ▶ ... which is great if capital is taxed today!
- ▶ Rationalizes why the planner likes an *positive* slope for capital taxes!
- ▶ Reverse holds for $IES > 1$: tax *decreases* to zero

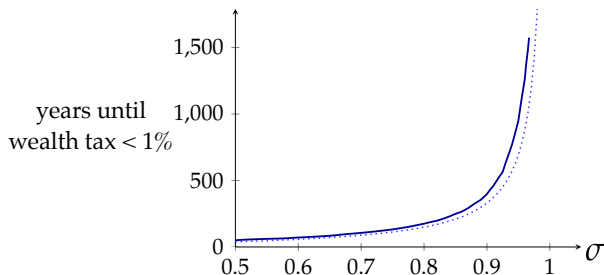
Capital and taxes for various IES's

- ▶ One can solve a recursive version of the planning problem
- ▶ Here: take $\gamma = 0$ and let σ range from 0.75 to 1.25
- ▶ Left graph: capital stock k_t , right graph: wealth tax τ_t



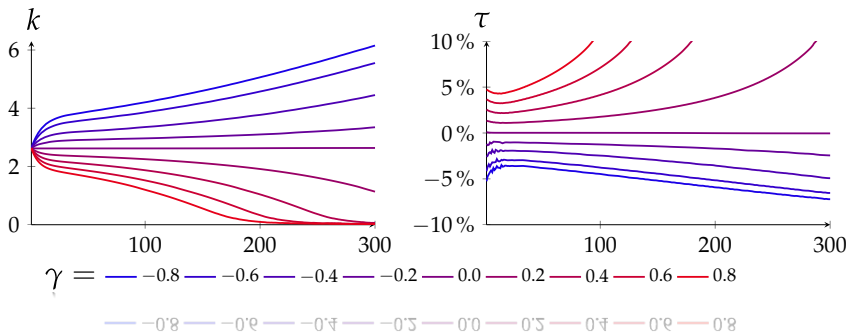
Slow convergence for $\sigma < 1$ (IES > 1)

- For $\sigma < 1$, tax does converge zero, but **convergence is slow**:



Capital and taxes for various degrees of redistribution

- ▶ Solve same planning problem, but now keep σ fixed at 1.25 and vary γ
 - ▶ normalize γ such that zero = no redistribution at zero tax steady state
 - ▶ $\gamma < 0 \rightarrow$ redistribution towards capitalists
 - ▶ $\gamma > 0 \rightarrow$ redistribution towards workers
- ▶ Illustrates that Prop 2 and 3 are robust to nonzero γ
- ▶ A more formal result is in Werning-Straub



General savings functions

- ▶ Proposition 2 can be generalized to almost arbitrary savings functions for capitalists
- ▶ Again assume $\gamma = 0$ for simplicity
- ▶ Suppose having time t income I_t , capitalists save exactly $S(I_t, R_{t+1}, R_{t+2}, \dots)$
 - ▶ naturally depends on future interest rates $\{R_{t+1}, R_{t+2}, \dots\}$
 - ▶ Assume S increases in I_t and weakly decreases in future interest rates
- ▶ **Proposition 4: S-W** Optimal tax rates **cannot** converge to zero (or anything negative)

Binding bounds

- ▶ in Chamley (1986) we found that period 1 tax can be large. So set upper bound on taxes
- ▶ If bounds on capital tax rates are asymptotically slack, then long run capital tax is zero.
- ▶ Was it reasonable to assume capital tax bounds do not bind indefinitely in Chamley (1986)?
- ▶ Move to continuous time (for simple bang-bang optimal tax policies)
- ▶ Assume separable isoelastic utility

$$\int_0^{\infty} e^{-\rho t} u(c_t, n_t) dt \quad u(c, n) = \frac{c^{1-\sigma}}{1-\sigma} - \frac{n^{1+\zeta}}{1+\zeta}$$

- ▶ Resource constraint

$$c_t + g + \dot{k}_t \leq f(k_t, n_t) - \delta k_t$$

- ▶ Budget constraint and implementability as before

Binding bounds

- ▶ in Chamley (1986) we found that period 1 tax can be large. So set upper bound on taxes
- ▶ If bounds on capital tax rates are asymptotically slack, then long run capital tax is zero.
- ▶ Was it reasonable to assume capital tax bounds do not bind indefinitely in Chamley (1986)?
- ▶ Move to continuous time (for simple bang-bang optimal tax policies)
- ▶ Assume separable isoelastic utility

$$\int_0^{\infty} e^{-\rho t} u(c_t, n_t) dt \quad u(c, n) = \frac{c^{1-\sigma}}{1-\sigma} - \frac{n^{1+\zeta}}{1+\zeta}$$

- ▶ Resource constraint

$$c_t + g + \dot{k}_t \leq f(k_t, n_t) - \delta k_t$$

- ▶ Budget constraint and implementability as before

Planning problem

- Planner solves

$$\max \int_0^{\infty} e^{-\rho t} u(c_t, n_t) dt$$

subject to RC and IC

$$c_t + g + \dot{k}_t \leq f(k_t, n_t) - \delta k_t$$

$$\int_0^{\infty} e^{-\rho t} (u_{ct} c_t + u_{nt} n_t) = u_{c0}(k_0 + b_0)$$

and bounds on taxes (assume $\bar{\tau} = 1$ for simplicity)

$$\frac{\dot{c}_t}{c_t} = \frac{1}{\sigma}(r_t - \rho)$$

$$r_t = (1 - \tau_t)(f_k(k_t, n_t) - \delta)$$

$$\tau_t \leq \bar{\tau}$$

Planning problem

- Planner solves

$$\max \int_0^{\infty} e^{-\rho t} u(c_t, n_t) dt$$

subject to RC and IC

$$c_t + g + \dot{k}_t \leq f(k_t, n_t) - \delta k_t$$

$$\int_0^{\infty} e^{-\rho t} (u_{ct} c_t + u_{nt} n_t) = u_{c0}(k_0 + b_0)$$

and bounds on taxes (assume $\bar{\tau} = 1$ for simplicity)

$$\frac{\dot{c}_t}{c_t} \geq -\frac{\rho}{\sigma}$$

Chamley (1986, Theorem 2)

- ▶ **Chamley (1986, Theorem 2):** Suppose $\bar{\tau} = 1$. Then there exists a time $T < \infty$ such that
 - ▶ $\tau_t = \bar{\tau}$ for $t < T$
 - ▶ $\tau_t = 0$ for $t > T$
- ▶ Bang-bang due to continuous time
- ▶ But why can T not be infinite?
 - ▶ Chamley's (1986) proof: "The bounds cannot be binding forever or marginal utility would grow to infinity, which is absurd"
- ▶ Next: Nothing absurd here ...

Positive long run capital taxation

- ▶ **Proposition 7: S-W** Take $\bar{\tau} = 1$ and $\sigma > 1$. Fix initial capital k_0 . Then there exist $\underline{b} < \bar{b}$ such that
 - ▶ if $b_0 \in [\underline{b}, \bar{b}] \Rightarrow T = \infty$!
- ▶ **For sufficiently high levels of initial debt b_0 , the bounds on capital taxes bind forever!**
- ▶ Can construct specific analytically tractable examples (see Straub-Werning)

Proof idea

- ▶ Planning problem with current value multipliers

$$\max \int_0^{\infty} e^{-\rho t} u(c_t, n_t) dt$$

$$c_t + g + \dot{k}_t \leq f(k_t, n_t) - \delta k_t \quad (\lambda_t)$$

$$\int_0^{\infty} e^{-\rho t} (u_{ct} c_t + u_{nt} n_t) \geq u_{c0}(k_0 + b_0) \quad (\mu)$$

$$\dot{c}_t \geq -\frac{\rho}{\sigma} c_t \quad (\eta_t)$$

- ▶ Note: $b_0 \uparrow \Rightarrow$ gov. needs to tax more $\Rightarrow$ IC constraint tighter $\Rightarrow \mu \uparrow$
- ▶ In fact: As b_0 approaches highest feasible debt level $\bar{b}$, $\mu \nearrow +\infty$
- ▶ Now pick $\sigma > 1$ and suff. high b_0 (hence high μ), and prove $T = \infty$

Proof idea (2)

- ▶ Consider FOC for consumption

$$\dot{\eta}_t - \rho\eta_t = \eta_t \frac{\rho}{\sigma} + \lambda_t - (1 - \mu(\sigma - 1)) u_{ct}$$

where tax bound $\tau_t = \bar{\tau}$ binds if $\eta_t < 0$

- ▶ Note that if $T < \infty \Rightarrow \eta_t = \dot{\eta}_t = 0 \ \forall t > T$, implying for such t

$$\underbrace{\lambda_t}_{\geq 0} = \underbrace{(1 - \mu(\sigma - 1))}_{\text{possibly } < 0!} \underbrace{u_{ct}}_{> 0}$$

- ▶ **This is impossible if $\sigma > 1$ and μ sufficiently large!**
- ▶ Hence indefinite capital taxation, $T = \infty$, is optimal in those cases

Judd (1999)

- ▶ Representative agent model as in Chamley (1986)
- ▶ Does not assume convergence of allocation
 - ▶ Instead assumes multiplier Λ_t is bounded
- ▶ **Result:** long-run average tax on capital is zero
- ▶ **Intuition:** exploding consumption taxes are infinitely distortionary
- ▶ **But, are bounds on endogenous multiplier reasonable?**

Model

- Use same continuous time planning problem as for Chamley

$$\max \int_0^{\infty} e^{-\rho t} u(c_t, n_t) dt$$

$$c_t + g + \dot{k}_t \leq f(k_t, n_t) - \delta k_t$$

$$\int_0^{\infty} e^{-\rho t} (u_{ct} c_t + u_{nt} n_t) = u_{c0}(k_0 + b_0)$$

$$\frac{\dot{c}_t}{c_t} \geq -\frac{\rho}{\sigma}$$

- Call $e^{-\rho t} u_{ct} \Lambda_t$ the multiplier on the resource constraint

Judd (1999)

- ▶ Planner's first order condition

$$\frac{\dot{\Lambda}_t}{\Lambda_t} = r_t - r_t^*$$

- ▶ If Λ_t converges: zero tax!

- ▶ **Judd (1999):** If there are $0 < \underline{\Lambda} < \bar{\Lambda}$ with $\Lambda_t \in [\underline{\Lambda}, \bar{\Lambda}]$, then average capital tax goes to zero,

$$\frac{1}{t} \int_0^t (r_s - r_s^*) ds \rightarrow 0$$

- ▶ Follows immediately from imposing the bounds on Λ_t !

- ▶ Are the bounds reasonable?

No, see the above positive tax result for Chamley: There, $\Lambda_t \rightarrow 0$

Judd (1999)

- ▶ Planner's first order condition

$$\frac{\dot{\Lambda}_t}{\Lambda_t} = r_t - r_t^*$$

- ▶ If Λ_t converges: zero tax!

- ▶ **Judd (1999):** If there are $0 < \underline{\Lambda} < \bar{\Lambda}$ with $\Lambda_t \in [\underline{\Lambda}, \bar{\Lambda}]$, then average capital tax goes to zero,

$$\frac{1}{t} \int_0^t (r_s - r_s^*) ds \rightarrow 0$$

- ▶ Follows immediately from imposing the bounds on Λ_t !

- ▶ Are the bounds reasonable?

No, see the above positive tax result for Chamley: There, $\Lambda_t \rightarrow 0$

Judd (1999): Alternative interpretation

- First order condition for capital implies

$$\text{MRS}_{t,t+s}^{\text{planner}} = \text{MRT}_{t,t+s} = \exp \left\{ - \int_0^s r_{t+\tilde{s}}^* d\tilde{s} \right\}$$

- Using the agent's Euler condition

$$\text{MRS}_{t,t+s}^{\text{planner}} = \text{MRS}_{t,t+s}^{\text{agent}} \frac{\Lambda_{t+s}}{\Lambda_t}$$

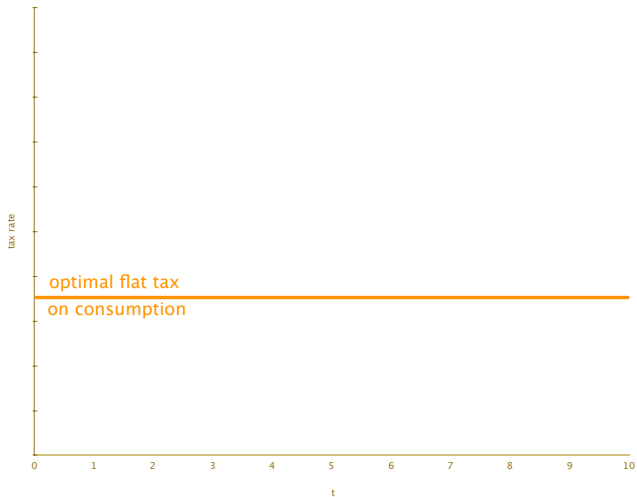
- Hence, **assuming Λ_t converges (or is bounded) is essentially assuming the result!**

Consumption tax intuition

- ▶ Common intuition for zero capital tax results:
 - ▶ an ever-rising tax on consumption is infinitely distortionary
 - ▶ hence not optimal
- ▶ **But:** here, there are **bounds on capital taxation**
 - ▶ this is **not** a standard Diamond-Mirrlees economy
 - ▶ the bounds force equivalent consumption taxes to be low initially...
 - ▶ ... which may make ever-rising consumption taxes a third-best

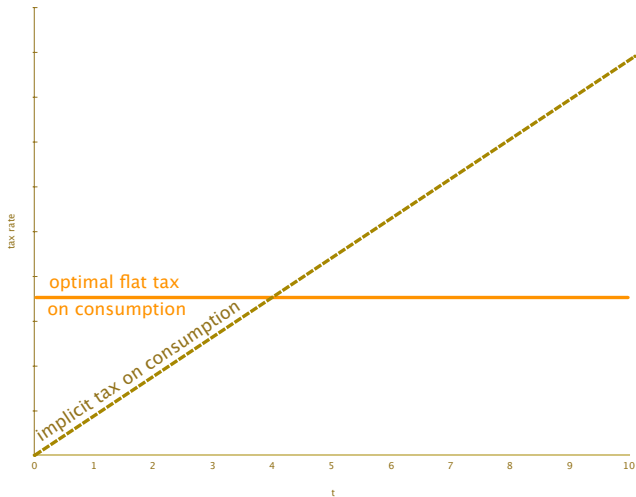
Graphical illustration

- **Flat** optimal consumption tax path without any capital tax bound



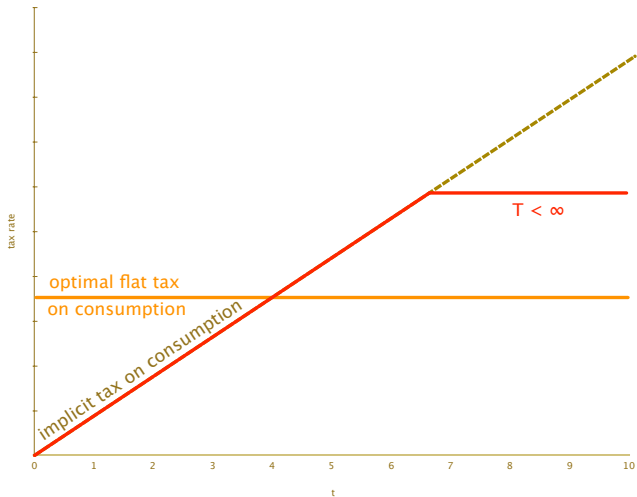
Graphical illustration

- Capital tax bound is equivalent to restriction on consumption taxes



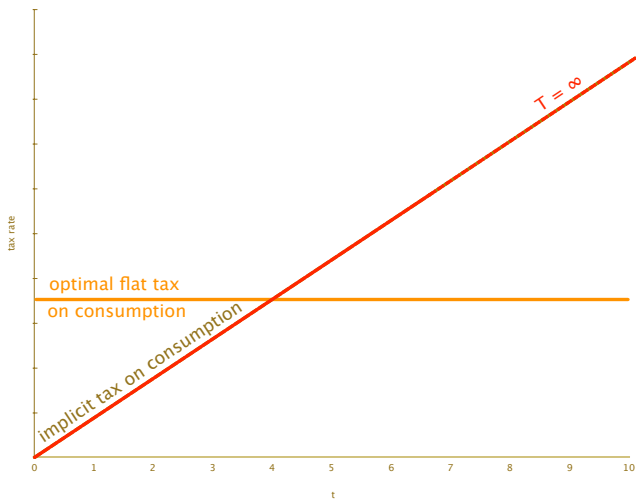
Graphical illustration

- For example, one could pick a consumption tax path like this...



Graphical illustration

- ... but it might well turn out that $T = \infty$ is actually optimal here



Consumption tax intuition more formally

- ▶ Straub-Werning adopt a simple linear technology framework to show that
 - ▶ yes, longer capital taxation creates larger distortions...
 - ▶ ...but indefinite capital taxation is **not** “infinitely distortionary”
- ▶ Hence there is no reason for why indefinite capital taxation cannot be optimal, *despite* an ever-increasing consumption tax!

Bibliography

Chamley, C. (1986). "Optimal Taxation of Capital Income in General Equilibrium with Infinite Lives." *Econometrica*, 54(3), 607–622.

Judd, K. L. (1985). "Redistributive Taxation in a Simple Perfect Foresight Model." *Journal of Public Economics*, 28(1), 59–83.

Judd, K. L. (1999). "Optimal Taxation and Spending in General Competitive Growth Models." *Journal of Public Economics*, 71(1), 1–26.

Straub, L., and Werning, I. (2020). "Positive Long-Run Capital Taxation: Chamley–Judd Revisited." *American Economic Review*, 110(1), 86–119. (Earlier version: NBER Working Paper No. 20441, 2014.)

Bibliography (cont.)

Lansing, K. J. (1999). "Optimal Redistributive Capital Taxation in a Neoclassical Growth Model." *Journal of Public Economics*, 73(3), 423–453.

Reinhorn, L. J. (2002). "On Optimal Redistributive Capital Taxation." Mimeo. (A later published version appears as Reinhorn (2019), *Journal of Public Economic Theory*, 21(3), 460–487.)

Goyal, R., Jensen, A., Lagakos, D., and Ndiaye, A. (2025). "Tax Productivity and Economic Development: A Quantitative Macroeconomic Analysis." Working paper.